S.no	Year	Authors	Dataset	Modality	Methodology	Accuracy	Identified Gaps	Citations
1.	2025	W. Zhai, S. Guo	35 participants	4-channel EEG + VR	Preprocessing + Feature Extraction + RF/SVM/KNN	86.67%	Lack of VR integration in existing systems; unclear channel selection and feature contribution	Zhai, W. and Guo, S. (2025) 'Anxiety status assessment method based on a portable EEG signal detection and Virtual Reality Integration System', *Proceedings of the 2025 3rd International Conference on Artificial Intelligence, Systems and Network Security*, pp. 279–284. doi:10.1145/3797161.3797205.
2.	2022	T. Wang, F. Li, X. Zhang, L. Huang, W. Jia	1320 EEG samples	20-channel EEG	Preprocessing + 1D-CNN + LSTM comparison	90.53%	Manual feature extraction in traditional ML is time-consuming and feature quality cannot be guaranteed; model and parameter optimization still needed	Wang, T. *et al.* (2022) 'A 1D-CNN prediction model for stroke classification based on EEG Signal', *Proceedings of the 8th International Conference on Communication and Information Processing*, pp. 191–196. doi:10.1145/3571662.3571695.
3.	2024	C. Lloyd, L. Lemoine, R. Al-Shaikh, K. T. Ly, H. Kayan, C. Perera, N. Pham	40 subjects	32-electrode EEG (SAM-40); adapted 16+6 channels for Neuro-GPT	Preprocessing + Neuro-GPT Fine-Tuning + ML Comparison	74.4%	Dataset lacked clear stress–relax distinction; self-reported stress ratings may be subjective; electrode mismatch between Neuro-GPT and SAM-40	Lloyd, C. *et al.* (2024) 'Stress-GPT: Stress detection with an EEG-based foundation model', *Proceedings of the 30th Annual International Conference on Mobile Computing and Networking*, pp. 2341–2346.

								doi:10.1145/3636534.3698121.
4.	2023	S. Raj, N. N. Sakhare, Y. K. Sharma, M. Soni, A. Bhange, P. K. Deshmukh	EEG Brainwave Dataset: Feeling Emotions	Multi-channel EEG	Preprocessing + CNN/LSTM/ConvLSTM + PSO	99.6%	Need for improved precision and efficiency; future scope includes multimodal integration for stronger emotion recognition	Raj, S. *et al.* (2023) 'Deep learningbased emotion detection from EEG signals for humanmachineinteraction', *Proceedings of the 5th International Conference on Information Management & Machine Intelligence*, pp. 1–6. doi:10.1145/3647444.3647930.
5.	2025	A. Kumar, A. Mathur, D. Joshi	18 participants	EEG (0.1–45 Hz, 200 Hz sampling)	FFT + PCA + SNN + XGBoost + SVC	60% (2-class), 43% (3-class)	Feature extraction approach may limit performance; need better real-time EEG representation to fully utilise SNN potential	Kumar, A., Mathur, A. and Joshi, D. (2025) 'Connecting brains and interfaces: Real-time EEG-based stress detection via spiking neural networks', *Proceedings of the ACM SIGCAS/SIGCHI Conference on Computing and Sustainable Societies*, pp. 666–671. doi:10.1145/3715335.3736311.
6.	2025	F. Zhu, B. Wu, Y. Huo, R. Dang, B. Hu, Q. Wang	42 adolescents	64-channel EEG + Eye Tracking	EVT + RGAT + MSNet Fusion	93.62%	High multimodal acquisition cost; limited dataset size; need simpler acquisition systems	Zhu, F. *et al.* (2025) 'MSNet: Multimodal self-attention network for depression detection via fusion of eye tracking and EEG', *Proceedings of the 2025 Symposium on Eye Tracking*

								Research and Applications, pp. 1–6. doi:10.1145/3715669.3723106.
7.	2025	Z. Yuan, Y. Yang, H. Liu	DEAP Dataset (32 subjects)	12-channel EEG	Nonlinear EEG Features + C4.5 Decision Tree	86.39%	Cross-modal fusion and complex emotion recognition remain challenging; need efficient real-time feedback and stronger privacy/safety mechanisms	Yuan, Z., Yang, Y. and Liu, H. (2025) 'Robotic recognition of human depression based on nonlinear EEG features', *Proceedings of the 2025 International Conference on Generative Artificial Intelligence for Business*, pp. 430–435. doi:10.1145/3766918.3766988.
8.	2024	Y. Xue, M. Zordan, X. Zhang, C. Chen	20 participants	Single-channel EEG (NeuroSky TGAM)	EEG Biofeedback + Stress Visualization	Not reported in the paper	Future work includes integrating additional biosignals and quantitative stress assessment	Xue, Y. *et al.* (2024) 'Minus X Minus = Plus: A semi-controlled data processing system based on EEG biofeedback for stress sharing', *Proceedings of the Twelfth International Symposium of Chinese CHI*, pp. 774–780. doi:10.1145/3758871.3758963.
9.	2025	A. Rezgui	CogWear Dataset (11 volunteers)	4-channel EEG + BVP + EDA + Temperature	FFT + Granger Causality + Stroop Test	Not reported in the paper	Causation with EDA and temperature was inconclusive; future work aims to identify stronger causation with other vital signs	Rezgui, A. (2025) 'On the causality between cognitive stress and physiological stress: The Stroop test as a case study', *Adjunct Proceedings of the 33rd ACM Conference on User Modeling, Adaptation and*

								Personalization, pp. 495–502. doi:10.1145/3708319.3734838.
10.	2025	Q. Liu, W. Ye, L. Zhang, Z. Liang	SEED, SEED-IV, DEAP	Multi-channel EEG	SCMM + Soft Contrastive Learning + Hybrid Masking	91.70%	Limited cross-corpus generalization in existing methods; reliance on traditional contrastive learning and domain adaptation reduces generalizability	Liu, Q. *et al.* (2025) 'EEG-SCMM: Soft contrastive masked modeling for cross-corpus EEG-based emotion recognition', *Proceedings of the 33rd ACM International Conference on Multimedia*, pp. 5834–5842. doi:10.1145/3746027.3755798.
11.	2021	Y. Wang, F. Liu, L. Yang	MODMA Dataset (53 subjects)	128-channel EEG	ITD + TCN (L-Softmax)	86.87%	CNN and RNN have limitations in temporal modeling and parallel computation; improved discriminative feature learning is needed	Wang, Y., Liu, F. and Yang, L. (2021) 'EEG-based depression recognition using intrinsic time-scale decomposition and temporal convolution network', *The Fifth International Conference on Biological Information and Biomedical Engineering*, pp. 1–6. doi:10.1145/3469678.3469683.
12.	2022	W. Zhang, W. Liu, X. Liu, S. Liu, D. Ming	63 subjects (30 MDD + 33 HC)	64-channel EEG (40Hz-ASSR)	ITC + TGC + SVM	81.8%	Objective biomarkers for MDD are still lacking; ASSR abnormalities in MDD remain underexplored; TGC in MDD has not been fully explored	Zhang, W. *et al.* (2022) 'Altered lateralization of gamma oscillations and theta-gamma coupling in major depression: An EEG study', *Proceedings of the 2022 9th International*

								Conference on Biomedical and Bioinformatics Engineering, pp. 203–208. doi:10.1145/3574198.3574230.
13.	2023	N. Sakib, T. Faruk, M. K. Islam	40 participants	14-channel wireless EEG	Frequency Filtering + Nonlinear Features + SVM	94.88%	GAD-7 based EEG screening and young-adult anxiety studies are limited	Sakib, N., Faruk, T. and Islam, M.K. (2023) 'Wireless EEG based anxiety screening among young adults using machine learning model', *Proceedings of the 2023 8th International Conference on Biomedical Imaging, Signal Processing*, pp. 97–103. doi:10.1145/3634875.3634889.
14.	2023	C. Liang, Y. Ju, X. Hao, Y. Zhang, S. Liu	20 subjects (10 GAD + 10 HC)	64-channel resting-state EEG	Power Spectrum + KNN/LDA/SVM	69.76%	Anxiety subtypes not differentiated; depression comorbidity not considered; small sample size; only abnormal channels analyzed; future work suggested broader features and brain-wide analysis	Liang, C. *et al.* (2023) 'Abnormal oscillations in beta and gamma of generalized anxiety disorder patients based on resting state EEG', *Proceedings of the 2023 10th International Conference on Biomedical and Bioinformatics Engineering*, pp. 201–207. doi:10.1145/3637732.3637741.
15.	2025	H. Liu, Y. Hu	SEED Dataset (8 subjects tested)	Multi-channel EEG	PSD + Improved EEG-TCNet + SVM	81.08%	Existing EEG emotion recognition methods still require improved signal recognition accuracy	Liu, H. and Hu, Y. (2025) 'EEG emotion signal recognition method based on improved EEG-tcnet', *Proceedings of the 2025 3rd International*

								Conference on Internet of Things and Cloud Computing Technology, pp. 302–307. doi:10.1145/3776865.3776914.
16.	2021	Shikha, M. Agrawal, M. A. Anwar, D. Sethia	DASPS Dataset (23 participants)	14-channel wireless EEG	Feature Extraction + RFECV + SSAE + Decision Tree	83.93%	Existing anxiety classification methods require improved accuracy and reduced feature redundancy	Shikha *et al.* (2021) 'Stacked sparse Autoencoder and machine learning based anxiety classification using EEG signals', *The First International Conference on AI-ML-Systems*, pp. 1–7. doi:10.1145/3486001.3486227.
17.	2025	S. Zhang, Y. Hu, X. Yi, S. Nanayakkara, X. Chen	Multimodal physiological data	EEG + Text + Images	IntervEEG-LLM + Multimodal Modeling + Mental Health Intervention	Not reported in the paper	EEG-based LLM intervention remains underexplored; real-world deployment, privacy, and additional physiological integration require further study	Zhang, S. *et al.* (2025) 'Interveeg-LLM: Exploring EEG-based multimodal data for customized mental health interventions', *Companion Proceedings of the ACM on Web Conference 2025*, pp. 2320–2326. doi:10.1145/3701716.3717550.